Summary and Minutes of the Chemical Engineering Advisory Board Meeting, 9-10 April 2026

The advisory board was asked to comment on various aspects of the curriculum, the meeting content, and any other issues that they would like to raise. The survey questions are underlined and in bold font below, followed by responses of individual members. A summary in red font appears at the end of each section.

SUMMARY AND R&A:

The general themes seen in the comments are:

- Academic advising needs improvement. COL Hill mentioned specifically he was surprised to hear cadets were not meeting with advisers. Cadets want more advice on curriculum options and on career planning.
- Several board members want to remove MC300 from the curriculum. Either MC300 is too easy, or it is not relevant to the chemical engineering profession.
- Board members want us to sustain meetings with cadets. This was very popular. However, several mentioned that they want an informal interaction. (Dr. Biaglow thinks the lunch with cadets was missed by the board and proposes a cadet-faculty-board mixer at the Firstie club on Thursday night).
- Two board members mentioned “School of PE” as an alternative to PPI.

Comment highlights:

“You have a GREAT program! You are attracting some of our academy's best and brightest. Keep up the great work, and please reach out if I can do anything in our ME courses to improve your programs. GO ARMY!”

“Thank you for inviting me. I find it to be an enriching/refreshing experience. You have an extremely well-rounded program.”

“The USMA Chemical Engineering Program takes its mission seriously, designing and carrying out a challenging, exciting (despite the tedium of working too hard every day), and rather exacting curriculum for the cadets. America should be proud of the work you do. I know I am.”

BOARD RESPONSES TO QUESTION 1:

<u>Based on the assessment data or on your personal opinion, is there a course that the program should add to the curriculum?</u>

Armstrong: Continue Mathematica course. Is there a way to add EE301 content into the process controls course and ditch EE301? Is there a way to do CH400 all year round? Maybe as part of other courses. Numerical methods with machine learning and AI..

Daniel: They want more hands-on in early classes. They need more familiarity with equipment before using it. CHEMCAD exposure sooner. Loved heat transfer labs in that class. Do that with others.

CHEMCAD “experiments” not just make solutions. Bring in guest lectures for career perspective. Make material science required.

DeForest: Drop the statics course to reduce workload and find a class in CH400 to dedicate to statics. Regarding CH400: replace with mandatory “School of PE”-type program. Generate a flow diagram showing how principles/classes connect to build future courses.

Dietrich: Feedback from cadet was to consider online PE prep course or integrate it as it was better than FE Prep course. Expensive $1200 per.

Hair: Some of the other board members have suggested adding material science to the curriculum. I agree with this. I also think that the statics course could be removed and that material digested elsewhere..

Hill: In our conversations with cadets, there was a sentiment that Material Science should be a required course instead of an elective. It seems like that could serve as a bridge between chemistry and real-world warfighting systems (good Army applications).

Krishnamoorthy: An “Intro to ChE” course during the 1st year prior to Mass & Energy Bal. This course could be a “broad” intro to ChE concepts without diving into the details. Topics could include Basic Excel Python, intro to CHEMCAD flowsheets, interesting case studies.

Liberatore: Curriculum is sound. One more chemistry course. One more ChE lab course (most programs have UO1 and UO2). Move MEB to Fall 2nd year, change the Chem2 as co-req not pre-req. Cut statics (MC300) ⟶ make another elective. Do you need Physics 2 and EE301? Lots of overlap for ChE needs.

Schultz: Incorporating introductions to the unit op lab equipment within the appropriate courses. This could allow them to do deeper analysis in ChE lab instead of worrying about running the equipment.

Shipe: Not necessarily a new course, but more a restructuring of the timing of courses. Most cadets had issues with not being introduced to any actual ChemE courses until it was too late to decide if ChemE was right for them. Also, there seems to be a lack of clarity among some cadets about the purpose of the labs in their earlier courses, and would rather see the labs/stations they get introduced to in CH459 introduced earlier, so it isn't their first time seeing the station and they can get into the actual coursework faster.

Tanev: The department has done a great job of revisiting this question regularly, and adjusted the curriculum to address specific points we’ve addressed in the past. One point that was brought up on electives was that they may be too easy for ChEs (i.e. MC300) – but looks like that’s already being addressed.

Theising: No, but based on cadet feedback, consideration could be given to earlier introduction to unit ops lab equipment where it fits within the context of other courses. Cadets also claimed other departments use FEE reference manual in course work which has helped them in FEE prep for those topics.

BOARD RESPONSES TO QUESTION 2:

Do you have any suggestions to improve the advisory board meeting for next year?

Armstrong: This format is working. The small group breakout sessions [were] good.

Daniel: No comments.

DeForest: I liked rotation with cadets from previous years and would prefer to keep it.

Dietrich: In person – remote not very effective.

Hair: I was unable to attend in person this year, sadly. However, the small group breakout sessions seems like an excellent way to hash out ideas in a less formal fashion. And I think that having lunch with the cadets, as we have sometimes done, would be an excellent way to get a flavor for their outlooks in a one-to-one fashion..

Hill: No. Once again, this was great. I appreciate the opportunity to engage with cadets and faculty. Coordination was fantastic.

Krishnamoorthy: None. Extremely well-organized!

Liberatore: Compare ChE with ME and Civil for total credit hours (with other programs on number of ChE courses.) Statics course is a great place to learn how LLMs work. Is this topic covered? 5th math course. Most ChE only [take] 4 (calc 1, 2, 3, diff eqn.).

Schultz: More informal interactions with the cadets where possible.

Shipe: I'd like some more informal time with the cadets outside of the panel Q&A to answer questions they may have that could be more personal to either their ChemE experience, or questions regarding what comes next both in the military and beyond.

Tanev: It would be great to hear more about prior ChE students! What they're doing now, in ChE industry or elsewhere in academia.

Theising: Small group discussions were great, would sustain that.

BOARD RESPONSES TO QUESTION 3:

Please add any additional comments that you would like to make below.

Armstrong: Long term ABET process. This is a heavy lift. Why losing military faculty early? CH459: another heavy lift. Could we implement 2 instructors again? Expand to 3 sections. The CAP on majors seems heavy handed; I like the idea of overage as we know 4-8 will drop before graduation.

Daniel: They recommend all take "Practice School of FE" $1200 but guaranteed to pass exam. Need better advising & shepherding through program, electives, professional goals. Advising needs more structure and improvement. Missed opportunity for expectation setting, networking, goal setting, elective advice, career pathways development, etc.

DeForest: As major population grows I would use Δ National FEE Avg for a KPI. How do we maintain the high standard with a larger population? I think CH400 approach (the current way) will need to be revamped to accommodate larger group. Familiarity with unit equipment lesson. Better introduction to CHEMCAD. Connect ChemE grads to Cows/Plebes about contemporary industry or relevance of Chem. Engineering.

Dietrich: More time with smaller cadet group.

Hair: The USMA Chemical Engineering Program takes its mission seriously, designing and carrying out a challenging, exciting (despite the tedium of working too hard every day), and rather exacting curriculum for the cadets. America should be proud of the work you do. I know I am.

Hill: Cadets mentioned it would be great if your courses used the FE reference manual in courses versus course-created reference manuals. Cadets also mentioned they could use more guidance from advisors (DACs). It was surprising to hear that some of the Firsties have never met with their DAC and were left to pick electives with no guidance whatsoever. They also asked for more exposure to industry and/or guest lectures.

You have a GREAT program! You are attracting some of our academy's best and brightest. Keep up the great work, and please reach out if I can do anything in our ME courses to improve your programs. GO ARMY!

Krishnamoorthy: Thank you for inviting me. I find it to be an enriching/refreshing experience. You have an extremely well-rounded program.

Liberatore: I will send small group notes via email.

Schultz: A holistic analysis of the timing of the courses in the academic sequence might help. The cadets would like to see MEB earlier if possible. More involvement of academic advisors with all students, not just proactive students would be good.

Shipe: I liked the small group discussions this year. It allowed for more focused and nuanced conversations about not just the cadet development, but faculty as well. Based on our group's conversation, there could be some benefit to having a source of periodic feedback for new instructors to develop them not just as instructors, but as leaders and help them determine what needs to be done to possibly become course directors.

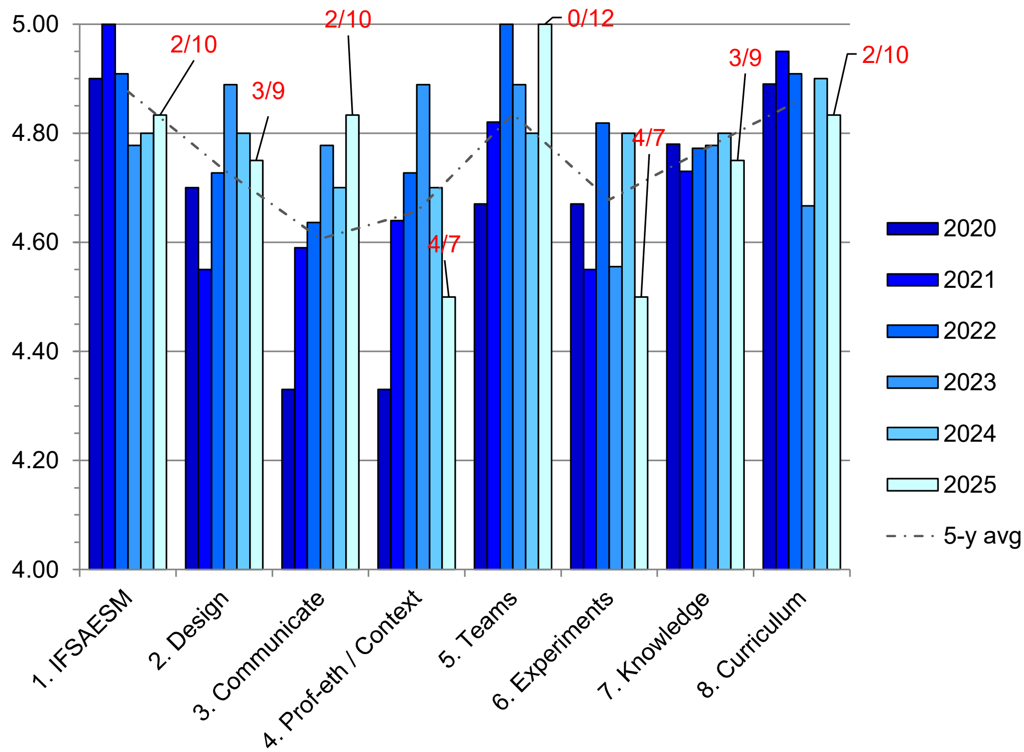
Tanev: The small group discussions were a great change for this year. In this quickly changing environment, it's important for students to be introduced to multiple programs and taught how to learn vs. specific programs, particularly given army commitment post-graduation. The department does a great job of this.

Theising: None, no comments. There were largely positive reviews on CH400. Adjustments the program has made and some of the early response in outcomes/test scores is awesome and encouraging. In two cases it was mentioned "daily quizzes" had resulted in improved results. I know the cadets would hate me for suggesting it and I also realize their limited bandwidth must be protected, but it makes me wonder if broader use of that method could be helpful. I know the department was avoiding an enrollment cap, but I think more good will come of it than bad. In my opinion a screening for candidates with a lower chance of success has been needed for a long time.

SUMMARY OF SURVEY RESULTS

Advisory Board Student Outcomes (SO) Survey Results:

The chemical engineering advisory board is asked to rate performance of cadets on student outcomes (SOs) based on data presented to the board at the advisory board meetings. Advisory board responses for AY2020 to AY2025 are shown in the figure below, including data from the most recent advisory board meeting, conducted 9-10 April 2026. Data for AY2026 is not available until after the advisory board meeting in spring of 2027. Data labels are response frequencies for responses of 4 or 5 (# of 4s / # of 5s) on the 1-5 Likert scale used in the survey. For example, in outcome 1 IFSAESM, 10 out of 12 board members responded with a 5 and there were two 4's, so the label is 2/10. The five-year average is the dotted line. The survey questions are below the figure.



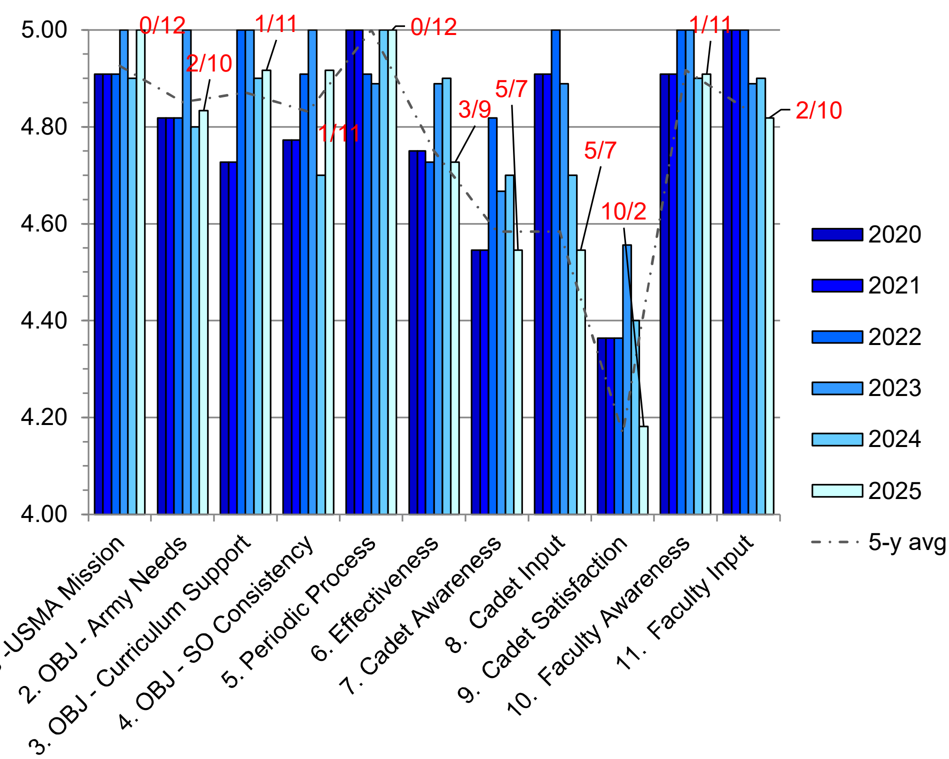
Survey Questions:

1. The cadets in the program are able to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. The cadets in the program are able to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. The cadets in the program are able to communicate effectively with a range of audiences.
4. The cadets in the program are able to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

5. The cadets in the program are able to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. The cadets in the program are able to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.
7. The cadets in the program are able to acquire and apply new knowledge as needed, using appropriate learning strategies.
8. The cadets in the program have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.

Advisory Board Program Educational Objectives (PEO) Survey Results:

The primary task of the advisory board is to assess the program educational objections (PEOs) of the chemical engineering program. A survey was administered to the board after a series of targeted activities involving the cadets and after a presentation of the PEOs by the program director. Advisory board responses to the program survey for AY2020 to AY2025 are shown in the figure below, including data from the most recent advisory board meeting conducted on 9-10 April 2026. As before, data for AY2026 (this year) will not be available until after the advisory board meeting in spring of 2027. Data labels are response frequencies for responses of 4 or 5 (# of 4s / # of 5s) on the 1-5 Likert scale used in the survey. For example, in question 1, which pertains to the consistency of the PEOs with the USMA mission, 12 out of 12 board members responded with a 5 and there were no 4's, so the label is 0/12. The five-year average is the dotted line. The survey questions are below the figure. A single relative low is seen in question 9, dealing with cadet satisfaction with the courses in the program.



Survey Questions:

1. The program objectives are consistent with the USMA mission.
2. The program objectives are consistent with the needs of the Army.
3. The program curriculum supports the program objectives.
4. The student outcomes are consistent with the program mission and objectives.
5. The program has a process for periodically assessing the achievement of its student outcomes.
6. The survey methods used by the program are effective.
7. The cadets in the program are aware of the program objectives.
8. The cadets are given an opportunity to provide their opinion about the program objectives.
9. The cadets are satisfied with the courses in the program.
10. The faculty are aware of the program objectives.
11. The faculty are given an opportunity to provide their opinion about the program objectives.